

HATTIESBURG, MS, USA

WMO: 747590

Lat: 31.265N

Lon: 89.253W

Elev: 45

StdP: 100.79

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 13833

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.5	-1.9	-11.1	1.5	2.7	-8.4	1.8	4.0	8.9	12.4	8.2	13.5	1.4	0	0.364

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	35.6	24.4	34.4	24.4	33.5	24.3	26.5	31.9	26.0	31.3	25.6	30.8	2.6	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.4	28.5	24.6	19.7	28.1	24.1	19.1	27.7	82.5	31.8	80.4	31.4	78.6	31.0	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	7.1	5.8	-6.8	36.9	2.2	1.8	-8.4	38.2	-9.7	39.2	-10.9	40.2	-12.5	41.5		
(5)			-7.6	27.4	2.2	0.7	-9.1	27.9	-10.4	28.3	-11.6	28.7	-13.2	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.5	9.6	11.8	15.7	19.3	23.3	27.0	27.8	27.8	25.5	20.0	14.1	11.2		
	DBStd	7.68	5.80	5.41	4.85	3.75	3.17	1.84	1.55	1.78	2.48	4.41	4.67	5.46		
	HDD10.0	196	80	38	10	1	0	0	0	0	0	1	14	52		
	HDD18.3	1022	275	191	108	33	6	0	0	0	1	34	143	231		
	CDD10.0	3652	68	90	188	279	412	509	551	551	466	312	136	89		
	CDD18.3	1435	4	9	28	62	159	259	292	293	217	87	15	9		
	CDH23.3	13638	11	46	202	559	1492	2612	2929	2946	1922	776	111	33		
	CDH26.7	5855	0	3	29	126	594	1244	1380	1396	820	249	12	1		
(14) Wind	WSAvg	2.1	2.5	2.7	2.7	2.7	2.2	1.8	1.6	1.6	1.7	1.8	2.1	2.4		
(15) Precipitation	PrecAvg	1556	159	146	153	126	123	120	139	135	103	84	119	144		
	PrecMax	2224	347	420	348	438	295	399	363	351	366	349	320	376		
	PrecMin	949	26	18	32	15	13	18	40	5	8	0	19	41		
	PrecStd	274	78	87	69	92	81	77	66	73	81	67	67	79		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.7	26.9	28.9	31.1	34.1	37.1	36.7	37.0	35.0	32.8	28.3	26.2		
		MCWB	18.5	19.7	19.2	21.8	22.7	23.7	24.0	25.1	23.6	21.9	20.5	20.7		
	2%	DB	22.9	24.8	27.5	29.4	32.6	35.1	35.0	35.3	33.7	31.0	26.3	24.1		
		MCWB	18.3	19.0	19.1	20.4	22.6	23.7	24.7	25.2	23.9	22.1	19.6	20.1		
	5%	DB	21.4	22.7	25.9	28.1	31.4	33.8	33.9	34.0	32.6	29.4	24.3	22.4		
		MCWB	17.5	17.7	18.2	19.9	22.2	23.9	24.8	25.0	23.8	21.4	18.5	19.2		
	10%	DB	19.3	21.1	23.8	26.8	30.2	32.6	32.8	32.9	31.3	27.8	22.7	20.5		
		MCWB	16.1	17.2	17.7	19.1	21.6	23.8	24.8	24.9	23.6	20.6	18.2	18.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.7	21.7	21.9	23.7	25.1	26.6	26.9	27.4	26.2	25.4	22.9	22.5		
		MCDB	22.8	24.1	25.0	27.9	30.3	31.4	32.3	32.6	30.9	28.8	25.5	24.5		
	2%	WB	19.6	20.6	20.8	22.4	24.1	25.8	26.3	26.7	25.5	24.4	21.7	20.9		
		MCDB	21.7	23.4	24.1	26.4	28.9	31.0	32.1	32.1	30.3	28.1	24.1	23.1		
	5%	WB	18.5	19.4	20.0	21.7	23.6	25.4	25.8	26.1	25.1	23.5	20.5	19.8		
		MCDB	20.8	21.9	23.4	25.6	28.4	30.7	31.3	31.5	29.8	27.2	23.2	21.9		
	10%	WB	16.7	17.8	19.1	20.9	23.0	24.9	25.4	25.6	24.6	22.6	19.1	18.2		
		MCDB	18.8	20.3	22.7	24.6	27.7	30.1	30.7	30.8	29.0	25.9	21.9	20.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	12.2	13.0	12.9	12.1	10.9	10.3	10.5	11.2	13.0	13.8	11.9		
		MCDBR	12.4	13.0	13.3	13.3	13.0	12.5	11.6	11.6	12.3	13.6	14.5	11.9		
		MCWBR	8.1	7.6	6.1	5.8	4.5	3.7	3.6	3.7	4.2	5.7	8.3	8.0		
		MCDBR	9.9	10.1	10.3	10.5	10.6	10.3	10.3	10.2	9.9	10.1	11.5	10.1		
	5% WB	MCWBR	7.4	7.1	5.8	5.8	4.2	3.8	3.7	3.8	3.8	4.9	7.8	7.8		
(40) Clear-Sky Solar Irradiance	taub		0.329	0.343	0.368	0.402	0.433	0.458	0.520	0.504	0.449	0.378	0.347	0.332		
	taud		2.526	2.483	2.416	2.332	2.275	2.251	2.099	2.142	2.265	2.440	2.475	2.527		
	Ebn at Noon		901	918	914	891	861	834	782	790	822	867	869	877		
	Edh at Noon		86	98	113	127	136	139	161	152	130	102	90	82		
(44) All-Sky Solar Radiation	RadAvg		2.85	3.41	4.54	5.64	6.18	6.16	5.88	5.60	5.09	4.35	3.28	2.52		
	RadStd		0.25	0.48	0.37	0.45	0.37	0.47	0.36	0.39	0.36	0.52	0.30	0.25		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (25 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	+136	+91

Nomenclature: See separate page